

Theorem and Corollary Summary for RandNLA Algorithm 2 Row Sampling

LeanFpAnalysis repository

Overview

This note summarizes the corrected Lean formalization for Algorithm 2, the row-sampling meta-algorithm from Drineas and Mahoney’s CACM RandNLA survey “[RandNLA: Randomized Numerical Linear Algebra](#)”. Algorithm 2 samples rows independently and returns an $s \times n$ matrix $\tilde{A}$. The relevant randomized object is the Gram matrix $\tilde{A}^T \tilde{A}$, not a grouped sum over repeated row hits.

The formalization uses the equation (4) distribution

$$p_i = \frac{\|A_{i*}\|_2^2}{\|A\|_F^2} = \frac{\sum_j A_{ij}^2}{\sum_{k,l} A_{kl}^2}.$$

If output row t samples input row i_t , then

$$\tilde{A}_{tj} = \frac{A_{i_t j}}{\sqrt{s p_{i_t}}}.$$

Row Distribution

Definition 1 (Norm-squared row probabilities). For $A \in \mathbb{R}^{m \times n}$, define

$$r_i = \sum_j A_{ij}^2, \quad D = \|A\|_F^2 = \sum_i r_i, \quad p_i = r_i / D.$$

Lean reference.

`rowNormSq, rowSqNormProbDen, rowSqNormProb.`

Theorem 2 (Probability normalization). *If $D \neq 0$, then $\sum_i p_i = 1$. If $D > 0$, then $p_i \geq 0$ for all rows i . If row i contains a nonzero entry, then $p_i > 0$.*

Lean reference.

`rowSqNormProb_sum_eq_one,`
`rowSqNormProb_nonneg,`
`rowSqNormProb_pos_of_entry_ne_zero.`

Corollary 3 (Nonzero row-scaling denominator). *If $s > 0$ and $p_i > 0$, then $\sqrt{s p_i} \neq 0$.*

Lean reference.

`rowSampleScaleDen_ne_zero.`

Sampled Sketch Stability

Definition 4 (Exact and floating sampled rows). The exact sampled sketch entry is

$$\tilde{A}_{tj} = \frac{A_{itj}}{\sqrt{sp_{i_t}}}.$$

The floating-point sampled sketch entry is

$$\widehat{\tilde{A}}_{tj} = \text{fl} \left(\frac{A_{itj}}{\sqrt{sp_{i_t}}} \right).$$

Lean reference.

`rowSampleIncrement`, `fl_rowSampleIncrement`, `rowSampleSketch`, `fl_rowSampleSketch`.

Theorem 5 (One sampled entry is stable). *If $\sqrt{sp_{i_t}} \neq 0$, then*

$$\left| \widehat{\tilde{A}}_{tj} - \tilde{A}_{tj} \right| \leq u |\tilde{A}_{tj}|.$$

Lean reference.

`fl_rowSampleSketch_error_bound`.

Remark. No random hit counter appears in this theorem. Algorithm 2 returns repeated sampled rows as repeated output rows; it does not add them into a single output entry.

Gram Matrix Expectation

Definition 6 (Exact and sampled Gram matrices). The exact Gram matrix is

$$(A^T A)_{jk} = \sum_i A_{ij} A_{ik}.$$

For a sampled trace, the sampled Gram matrix is

$$(\tilde{A}^T \tilde{A})_{jk} = \sum_{t=1}^s \tilde{A}_{tj} \tilde{A}_{tk}.$$

Lean reference.

`rowGram`, `rowSketchGram`, `rowSampleGram`, `fl_rowSampleGram`, `fl_rowSampleGramDot`.

Theorem 7 (Elementwise unbiasedness of $\tilde{A}^T \tilde{A}$). *Assume $D > 0$ and $s > 0$. Under the independent norm-squared row sampler,*

$$\mathbb{E} \left[(\tilde{A}^T \tilde{A})_{jk} \right] = (A^T A)_{jk}$$

for every pair of columns j, k .

Lean reference.

`rowSqNormTraceProbability_expectationReal_rowSampleGram_entry`.

Equation (5)

Lemma 8 (Finite iid variance kernel). *For any scalar row statistic $f(i)$, if*

$$\mu = \sum_i p_i f(i),$$

then the independent trace satisfies

$$\mathbb{E} \left[\left(\frac{1}{s} \sum_{t=1}^s f(i_t) - \mu \right)^2 \right] = \frac{1}{s} \sum_i p_i (f(i) - \mu)^2.$$

This is proved from the product trace law, not assumed.

Lean reference.

`rowSqNormTraceProbability_expectationReal_sampleAverage_sub_mean_sq.`

Theorem 9 (Squared Frobenius second moment). *Assume $D > 0$ and $s > 0$. Then*

$$\mathbb{E} \left[\|\tilde{A}^T \tilde{A} - A^T A\|_F^2 \right] \leq \frac{D^2}{s}.$$

Since $D = \|A\|_F^2$, this is

$$\mathbb{E} \left[\|\tilde{A}^T \tilde{A} - A^T A\|_F^2 \right] \leq \frac{\|A\|_F^4}{s}.$$

Lean reference.

`rowSqNormTraceProbability_expectationReal_rowSampleGram_frob_error_sq_le.`

Corollary 10 (Expectation consequence). *Assume $D > 0$ and $s > 0$. Then*

$$\mathbb{E} \left[\|\tilde{A}^T \tilde{A} - A^T A\|_F \right] \leq \frac{1}{\sqrt{s}} \|A\|_F^2.$$

This expectation bound is useful, but it is weaker than the high-probability form of equation (5). The Lean proof derives it from the squared-moment theorem using a finite Cauchy–Schwarz/Jensen inequality proved in the repository’s finite-probability library.

Lean reference.

`FiniteProbability.expectationReal_le_sqrt_expectationReal_sq,`
`rowSqNormTraceProbability_expectationReal_rowSampleGram_frob_error_le.`

Theorem 11 (Equation (5), high-probability form). *Assume $D > 0$, $s > 0$, and $\varepsilon > 0$. Then*

$$\Pr \left[\|\tilde{A}^T \tilde{A} - A^T A\|_F \leq \varepsilon \|A\|_F^2 \right] \geq 1 - \frac{1}{s\varepsilon^2}.$$

Equivalently, if $1/(s\varepsilon^2) \leq \delta$, then the same event has probability at least $1 - \delta$. This is the high-probability Markov consequence of the squared Frobenius second moment above.

Lean reference.

`FiniteProbability.eventProb_le_ge_one_sub_expectationReal_sq_div`
`rowSqNormTraceProbability_eventProb_rowSampleGram_frob_error_le_epsilon`
`rowSqNormTraceProbability_eventProb_rowSampleGram_frob_error_le_epsilon_of_budget`

Floating-Point Effect on the Gram Matrix

Theorem 12 (Entrywise Gram perturbation from sampled-sketch stability). *Suppose a computed sampled sketch $\widehat{\tilde{A}}$ satisfies*

$$|\widehat{\tilde{A}}_{tj} - \tilde{A}_{tj}| \leq u|\tilde{A}_{tj}|$$

for every sampled output entry. Then each Gram entry satisfies

$$\left| (\widehat{\tilde{A}}^T \widehat{\tilde{A}})_{jk} - (\tilde{A}^T \tilde{A})_{jk} \right| \leq (2u + u^2) \sum_{t=1}^s |\tilde{A}_{tj}| |\tilde{A}_{tk}|.$$

Lean reference.

`rowSampleGram_entry_error_bound_of_entrywise.`

Corollary 13 (Expected Gram-entry bias from sampled-sketch stability). *Under the same entrywise stability assumption, the expected bias in each computed Gram entry is bounded by the expected deterministic perturbation:*

$$\left| \mathbb{E} \left[(\widehat{\tilde{A}}^T \widehat{\tilde{A}})_{jk} \right] - (A^T A)_{jk} \right| \leq (2u + u^2) \mathbb{E} \left[\sum_{t=1}^s |\tilde{A}_{tj}| |\tilde{A}_{tk}| \right].$$

For the concrete floating-point sketch, Lean states the same result with u instantiated as the model unit roundoff.

Lean reference.

`rowSqNormTraceProbability_expectationReal_rowSketchGram_entry_bias_bound_of_entrywise`
`rowSqNormTraceProbability_expectationReal_fl_rowSampleGram_entry_bias_bound_of_entrywise`

Corollary 14 (Expectation bound with a floating-point perturbation term). *For any perturbed sampled sketch B ,*

$$\mathbb{E} [\|B^T B - A^T A\|_F] \leq \frac{1}{\sqrt{s}} \|A\|_F^2 + \mathbb{E} [\|B^T B - \tilde{A}^T \tilde{A}\|_F].$$

In particular, taking $B = \widehat{\tilde{A}}$ gives the floating-point version: the exact RandNLA equation (5) term remains, and all rounding effects are isolated in the expected Gram perturbation $\mathbb{E}[\|\widehat{\tilde{A}}^T \widehat{\tilde{A}} - \tilde{A}^T \tilde{A}\|_F]$.

Lean reference.

`FiniteProbability.expectationReal_mono,`
`FiniteProbability.abs_expectationReal_le_expectationReal_abs.`

`rowSqNormTraceProbability_expectationReal_rowSketchGram_frob_error_le_add_perturb`
`rowSqNormTraceProbability_expectationReal_fl_rowSampleGram_frob_error_le_add_perturb`

Remark (High-probability perturbation transfer). Lean first proves a generic transfer lemma. If the floating-point Gram perturbation is known only with high probability,

$$\Pr \left[\|\widehat{\tilde{A}}^T \widehat{\tilde{A}} - \tilde{A}^T \tilde{A}\|_F \leq \tau \right] \geq 1 - \delta_\tau.$$

Then

$$\Pr \left[\|\widehat{\widehat{A}}^T \widehat{A} - A^T A\|_F \leq \varepsilon \|A\|_F^2 + \tau \right] \geq 1 - \left(\frac{1}{s\varepsilon^2} + \delta_\tau \right).$$

This is a union-bound transfer statement, not a new concentration theorem for the floating-point perturbation. It is useful only if one later proves a nontrivial high-probability perturbation bound with a smaller τ than the worst-case deterministic budget.

Corollary 15 (High-probability equation (5) with exact Gram summation). *Assume $D > 0$, $s > 0$, and $\varepsilon > 0$. If the sampled rows are rounded but the Gram matrix of the rounded sketch is formed as an exact sum, define*

$$\tau_{\text{scale}} = \left\| \left((2u + u^2)D \right)_{j,k=1}^n \right\|_F.$$

Then

$$\Pr \left[\|\widehat{\widehat{A}}^T \widehat{A} - A^T A\|_F \leq \varepsilon \|A\|_F^2 + \tau_{\text{scale}} \right] \geq 1 - \frac{1}{s\varepsilon^2}.$$

Lean proves that zero-probability sampled rows have zero mass under the product trace law; on the probability-one support, every sampled row has positive probability, so the floating-point division model applies to the row scaling. The Gram perturbation is then bounded by

$$\|(2u + u^2) \sum_{t=1}^s |\widetilde{A}_{tj}| |\widetilde{A}_{tk}|\|_F \leq \tau_*.$$

Thus no additional floating-point failure probability is paid: $\delta_\tau = 0$.

Corollary 16 (High-probability equation (5) with floating-point dot products). *Assume in addition the repository's dot-product guard `gammaValid fp s`. Let*

$$\tau_{\text{dot}} = \left\| \left(\gamma_s (1 + u)^2 D \right)_{j,k=1}^n \right\|_F, \quad \tau_{\text{full}} = \tau_{\text{scale}} + \tau_{\text{dot}}.$$

If each Gram entry is computed by the already formalized `fl_dotProduct` algorithm, then

$$\Pr \left[\|\text{fl_dot}(\widehat{\widehat{A}})^T \text{fl_dot}(\widehat{\widehat{A}}) - A^T A\|_F \leq \varepsilon \|A\|_F^2 + \tau_{\text{full}} \right] \geq 1 - \frac{1}{s\varepsilon^2}.$$

This is the fully floating-point corollary. The Gram dot-product error is not reproved inside the `RandNLA` development; Lean reuses the library theorem `dotProduct_error_bound`.

Lean reference.

`FiniteProbability.eventProb_inter_ge_one_sub_add.`

`rowSqNormTraceProbability_eventProb_fl_rowSampleGram_frob_error_le_epsilon_add_tau`

`rowSqNormTraceProbability_eventProb_fl_rowSampleGram_frob_error_le_epsilon_add_tau_of_forall`

`rowSqNormTraceProbability_eventProb_fl_rowSampleGram_frob_error_le_epsilon_add_tau_of_entrywise_budget`

`rowSqNormTraceProbability_eventProb_rowTracePositiveProb`

`rowSampleGramFpPerturbBudget`

`rowSampleGramDotProductBudget`

`rowSampleGramFullFpPerturbBudget`
`rowSampleGram_perturb_budget_le_explicit`
`rowSampleGram_dot_product_budget_le_explicit`
`rowSqNormTraceProbability_eventProb_fl_rowSampleGram_frob_error_le_epsilon_add_explicit_budget`
`rowSqNormTraceProbability_eventProb_fl_rowSampleGramDot_frob_error_le_epsilon_add_explicit_budget`

File Organization

- `LeanFpAnalysis/FP/Algorithms/RandNLA/RowSampling.lean` contains equation (4), row traces, sampled-row scaling, and local sampled-entry floating-point stability, plus the probability-one support theorem for positive-probability sampled rows.
- `LeanFpAnalysis/FP/Algorithms/RandNLA/RowSamplingGram.lean` contains the Gram matrices, independent-trace marginal facts, unbiasedness, the iid variance proof, high-probability equation (5), and the floating-point Gram perturbation theorems, including the exact-sum and fully floating-point explicit-budget FP corollaries.
- `LeanFpAnalysis/FP/Algorithms/DotProduct.lean` provides `fl_dotProduct` and `dotProduct_error_bound`; the fully floating-point Algorithm 2 Gram theorem reuses these results.
- `LeanFpAnalysis/FP/Analysis/FiniteProbability.lean` contains the reusable finite Markov, union-bound, monotonicity, absolute-expectation, and Cauchy–Schwarz/Jensen lemmas used by the expectation and high-probability arguments.